

Interference Effects of Different Resistance-Training Protocols on Rowing Ergometer Performance: A Study on Semiprofessional Rowers

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Purpose: To evaluate the interference effects of various resistance-training (RT) protocols on rowing ergometer performance. **Methods:** Fourteen semiprofessional male rowers randomly completed 5 protocols in separate sessions: (1) *control*—no RT session was performed, (2) *upper-body high-fatigue*—4 sets to failure during the bench pull exercise, (3) *upper-body low-fatigue*—4 sets of 6 repetitions during the bench pull exercise, (4) *lower-body high-fatigue*—4 sets to failure during the leg-press exercise, and (5) *lower-body low-fatigue*—4 sets of 6 repetitions during the leg-press exercise. All sets were performed against the 12-repetition-maximum load with 2 minutes of inter-set rest. Following the completion of the protocols, subjects performed an all-out 1000-m rowing ergometer test. **Results:** Compared with the control condition, rowing ergometer performance was not significantly affected after the low-fatigue RT protocols (upper body: $P \geq .487$; $\Delta = 0.0\% - 0.2\%$; lower body: $P \geq .200$; $\Delta = -0.2\% - 0.5\%$), while it significantly declined following high-fatigue RT protocols (upper body: $P \leq .001$; $\Delta = 1.0\% - 2.0\%$; lower body: $P \leq .002$; $\Delta = 2.1\% - 2.5\%$). The average heart rate was significantly lower for the control condition compared with all RT protocols ($P \leq .043$; $\Delta = 1.0\% - 1.5\%$). **Conclusions:** To minimize interference on rowing performance, coaches should prioritize the level of effort in RT protocols over specific exercises, specifically avoiding high-fatigue protocols that lead to failure before rowing practice.

Keywords: concurrent training, fatigue, sport, testing

Rowing is a highly demanding Olympic sport that requires exceptional strength and endurance.¹ A typical rowing competition involves covering a straight 2000-m course in the shortest time possible, with rowers performing over 200 strokes during a 5- to 7-minute race.^{2,3} The sport relies heavily on the coordinated efforts of the cardiorespiratory and muscular systems, with approximately 70% to 80% of energy derived from aerobic pathways and 20% to 30% from anaerobic metabolism.^{2,4} Upper- and lower-limb strength and power are crucial for rowing performance, particularly during the initial strokes of the race.⁵ As endurance capacity plays a critical role in maintaining high force levels throughout the race, effective training strategies must be implemented by strength and conditioning coaches to optimize both strength and endurance.

Concurrent training (CT), which involves the inclusion of both resistance training (RT) and endurance training (ET) within the same session, is widely used to enhance rowing performance.^{1,6,7} However, the training adaptations to CT depend on numerous factors such as the exercise type, volume, intensity, muscle groups targeted, training status, and training sequence. For instance, Murlasits et al⁸ reported that performing RT before ET leads to

superior adaptations in lower-body strength compared with ET followed by RT, whereas the training sequence had no impact on aerobic capacity. Furthermore, Petré et al⁹ suggested that the training status of individuals may affect CT adaptations, with lower-limb strength gains being negatively affected in trained individuals but not in moderately or untrained individuals. Interestingly, excessive training volume and inadequate recovery can also hinder subsequent performance and training adaptations due to interference between neuromuscular and metabolic processes.^{1,8,9} Considering the time constraints in elite sport settings,¹⁰ separating RT and ET into different days becomes challenging. Therefore, finding an optimal integration of both training modalities is crucial to maximize rowing performance.

Muscle fatigue during RT is characterized by a decline in the ability to generate force, and it is closely related to the number of repetitions completed in a set.^{6,11} Protocols that induce muscle failure have been found to elicit greater mechanical and metabolic stress, such as increased velocity loss, blood ammonia, lactate, cortisol, and creatine kinase levels, compared with nonfailure training.^{11,12} For example, Morán-Navarro et al¹³ reported significantly higher acute decline in velocity and jump height, as well as longer times of recovery following a RT protocol to muscle failure (3×10 against the 10-repetition maximum [10RM]) compared with lower levels of effort (3×5 and 6×5 repetitions against the 10RM load). Regarding long-term adaptations, Izquierdo-Gabarrén et al⁶ demonstrated that nonfailure RT including bench pull, seated cable row, lat pulldown, and power clean exercises

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resulted in greater gains in maximal strength, power output, and rowing performance compared with failure-based training in trained male rowers. However, it should be noted that successful elite rowers generate during the rowing stroke more power with their legs (75%–80%) than with their arms (20%–25%),¹⁴ suggesting that lower-body RT exercises may have a greater impact on rowing performance than upper-body exercises. It is also important to note that a meta-analysis has suggested that the interference effects between ET and RT are body part specific, as ET primarily lower affected lower- but not upper-body strength performance.⁷ Thus, it could be reasonable to assume that lower-body RT exercises may acutely impair rowing performance in a greater extent than upper-body RT exercises. However, the specific interference effects of different RT protocols on rowing performance remain unclear, especially regarding the comparison between lower- and upper-body exercises.

To address these research gaps, a group of semiprofessional rowers performed an all-out 1000-m rowing ergometer test immediately after completing 4 different RT protocols that differed in the level of effort (repetitions completed to failure [high-fatigue] vs half of the maximum possible number of repetitions [low-fatigue]) and muscle groups targeted in the RT protocol (bench pull [upper body] vs leg press [lower body]). Therefore, the main objective of the present study was to evaluate the level of interference on rowing ergometer performance induced by 4 different RT protocols: (1) upper-body high-fatigue, (2) upper-body low-fatigue, (3) lower-body high-fatigue, and (4) lower-body low-fatigue. We hypothesized that rowing ergometer performance would decline after completing high-fatigue RT protocols, while low-fatigue RT protocols would not cause systematic changes.^{1,6} Furthermore, it was hypothesized that lower-body RT protocols would have a greater negative impact on rowing performance compared with upper-body RT protocols.⁷

Methods

Participants

We conducted a sample size estimate with G*Power software (version 3.1.9.6) considering a conservative correlation coefficient (r) between repeated measures for rowing ergometer performance of .90 based on our pilot data. Considering an effect size of 0.25, alpha level of .05, power of 0.8, 2 groups, and 5 measurements, a sample size of 8 participants appeared to be necessary to detect the differences in rowing ergometer performance as a consequence of the fatigue induced by the different RT protocols. We conservatively recruited 14 highly active male competitive rowers (age = 17.1 [2.1] y [range: 16–25 y], body mass = 66.2 [9.0] kg, and body height = 1.74 [0.10] m). All rowers had prior experience competing in national events, and 4 of them had also taken part in international events. Additionally, 7 rowers were part of the Chilean National Promise Program developed by the Sports Institute. At the time of the study, participants were engaged in a rigorous training regimen that included 6 rowing sessions and 2 RT sessions per week. None of the rowers had any physical limitations that could compromise their performance during the tests implemented in this study. Prior to the commencement of the study, all participants were informed of the study procedures and they or their legal guardians (for participants aged <18 y) signed a written informed consent form. The study protocol adhered to the principles outlined in the Declaration of Helsinki and was approved by the institutional review board of the University of Granada.

Study Design

A crossover study design was used to elucidate whether 1000-m rowing ergometer performance could be compromised in competitive rowers immediately after completing different RT protocols. Over the course of 3 consecutive weeks, participants attended the laboratory on 6 separate occasions, with each session being separated by a minimum of 72 hours (see Figure 1 for the study timeline). The initial session was used to determine the 12RM for both the bench pull and leg press exercises. The 5 remaining experimental sessions were performed in a randomized and counterbalanced order. In each experimental session, immediately following completion of one of the designated RT protocols, a maximal 1000-m rowing ergometer test was performed. The RT protocols included the following: (1) *control*—no RT session was performed, (2) *upper-body high-fatigue*—4 sets to failure during the bench pull exercise with 2 minutes of interset rest against the 12RM load, (3) *upper-body low-fatigue*—4 sets of 6 repetitions during the bench pull exercise with 2 minutes of interset rest against the 12RM load, (4) *lower-body high-fatigue*—4 sets to failure during the leg press exercise with 2 minutes of interset rest against the 12RM load, and (5) *lower-body low-fatigue*—4 sets of 6 repetitions during the leg press exercise with 2 minutes of interset rest against the 12RM load. The 6 testing sessions were performed at the same time of the day for individual participants and under similar environmental conditions. Given the participants' extensive experience with the 2 RT exercises (leg press and bench pull) and the rowing ergometer, which are fundamental components of their regular training routines, a familiarization session was deemed unnecessary. Participants were instructed to refrain from engaging in strenuous physical activity within the 24 hours preceding each testing session.

Preliminary Session (Session 1)

Body mass (Ventus B-300A) and height (Avanutri 201) were measured at the beginning of the testing session. Following the anthropometric measurements, a standardized warm-up consisting of 5 minutes of jogging and joint mobilization exercises was completed. Subsequently, participants performed incremental loading tests to determine their 12RM load (ie, the load at which participants can perform a maximum of 12 repetitions) for the leg press and bench pull exercises. The initial load of the incremental loading tests was set at 50% of the self-reported 12RM. To avoid excessive fatigue, a skilled strength and conditioning coach closely supervised the participants, interrupting the set after completing 6 repetitions when it became evident that participants could exceed 12 repetitions with the given load. Afterward, the load was progressively increased by consensus between the coach and the rower until the load that allowed to perform a maximum of 12 repetitions was reached. Rest periods of 4 minutes were provided between successive sets. On average, participants required 5.2 (1.4) sets to reach the 12RM load for the leg press exercise and 4.9 (0.7) sets for the bench pull exercise. The 12RM load was 122.1 (36.2) kg (range: 70–186 kg) for the leg press and 54.1 (21.4) kg (range: 27–97 kg) for the bench pull exercise. The execution technique for both the leg press and bench pull exercises followed detailed guidelines provided elsewhere.^{15,16}

Experimental Sessions (Sessions 2–6)

The sequence of activities during the experimental sessions was as follows: (1) RT warm-up, (2) RT session, (3) rowing ergometer

Experimental protocols (sessions 2-6)

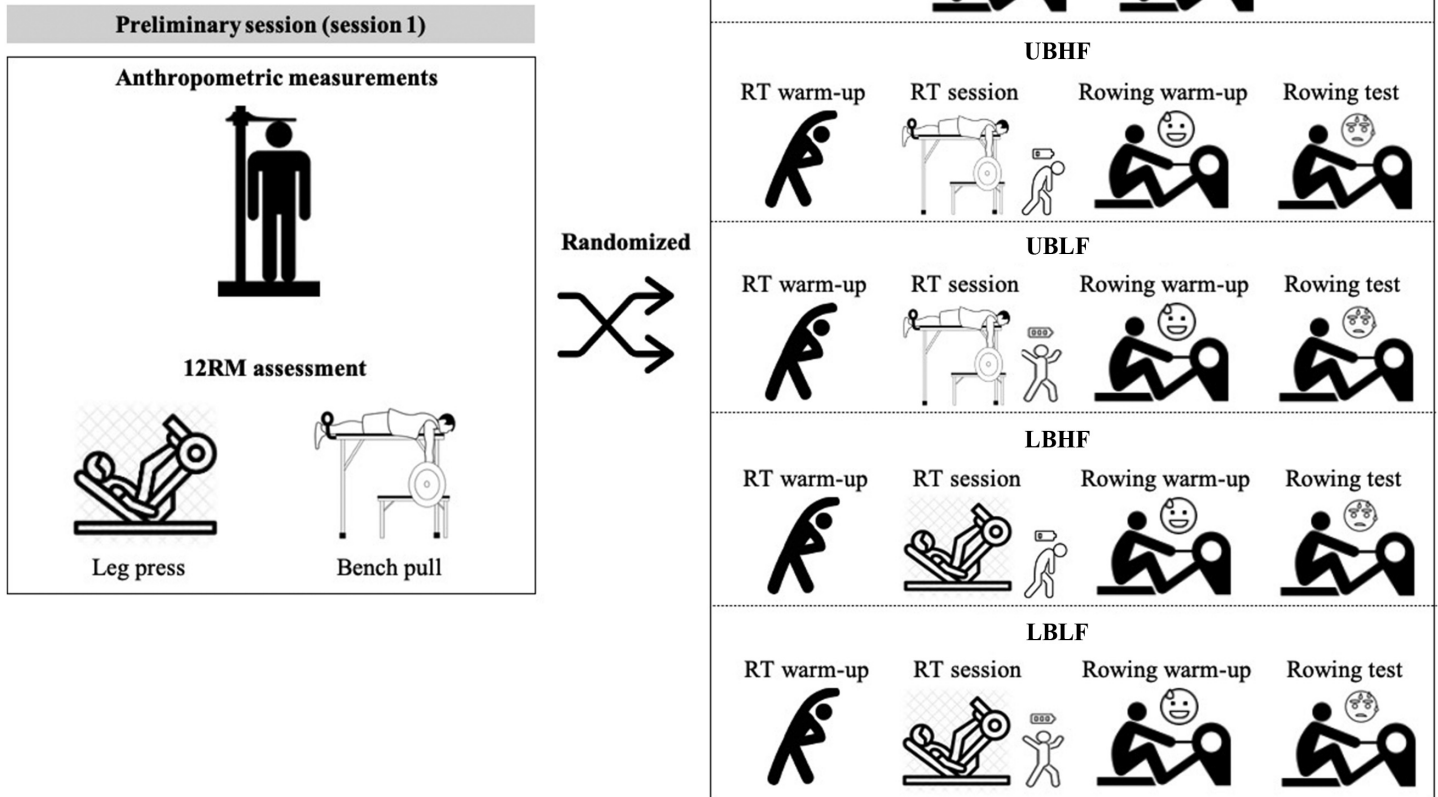


Figure 1 — Overview of the experimental design. The 5 experimental sessions differed in the RT protocol: (1) control—no RT session was performed, (2) UBHF—4 sets to failure during the bench pull exercise with 2 minutes of interset rest against the 12RM load, (3) UBLF—4 sets of 6 repetitions during the bench pull exercise with 2 minutes of interset rest against the 12RM load, (4) LBHF—4 sets to failure during the leg press exercise with 2 minutes of interset rest against the 12RM load, and (5) LBLF—4 sets of 6 repetitions during the leg press exercise with 2 minutes of interset rest against the 12RM load. 12RM indicates 12-repetition-maximum; LBHF, lower-body high-fatigue; LBLF, lower-body low-fatigue; UBHF, upper-body high-fatigue; UBLF, upper-body low-fatigue; RT, resistance training.

warm-up, and (4) 1000-m rowing ergometer test. Of note is that in the control condition, the testing session began directly from the rowing ergometer warm-up (RT warm-up and RT session were not performed).

RT Warm-Up

The warm-up consisted of 5 minutes of jogging and joint mobilization exercises, followed by 3 sets of the tested exercise (bench pull or leg press) separated by 2 minutes of rest. The first set comprised 12 repetitions at 50% of their 12RM, the second set involved 6 repetitions at 75% of their 12RM, and the third set entailed 3 repetitions at their 12RM. Once the warm-up was completed, participants were allowed to rest for 3 minutes before starting the first set of the RT session.

RT Session

Each RT session consisted of 4 sets performed at the 12RM load with 2 minutes of interset rest. The RT protocols differed in the

level of effort (repetitions completed to failure [high-fatigue] vs half of the maximum possible number of repetitions [low-fatigue]) and muscle groups involved in the RT protocol (bench pull [upper body] vs leg press [lower body]). The specific characteristics of the 4 RT protocols are as follows: (1) *upper-body high-fatigue*—bench pull exercise performed to failure, (2) *upper-body low-fatigue*—6 repetitions of the bench pull exercise, (3) *lower-body high-fatigue*—leg press exercise performed to failure, and (4) *lower-body low-fatigue*—6 repetitions of the leg press exercise. Once the last set of the RT session was performed, participants rested for 5 minutes before initiating the rowing ergometer warm-up procedure. The 5-minute recovery time was selected based on the rowers' common practice when incorporating both training modalities (ET and RT) within the same session.

Rowing Ergometer Warm-Up

The warm-up consisted of 5 minutes of full-body dynamic stretching exercises, 5 minutes of rowing at a pace boat of 130 seconds

for every 500 m (ie, it takes 130 s to complete a distance of 500 m), and 2 all-out starts of 8 rowing strokes separated by 15 seconds of rest. The time elapsed between the completion of the warm-up and the beginning of the 1000-m rowing ergometer test was set to 2 minutes.

1000-m Rowing Ergometer Test

Rowers were strongly encouraged by their coach to complete the 1000-m distance in the shortest time possible. According to the Chilean Rowing Federation, prior to each test the resistance lever on the rowing machine flywheel was set at 120 (no units). The following dependent variables were considered: time needed to complete 250 m (T250), 500 m (T500), and 1000 m (T1000) as well as the average heart rate (HR) reached during the 1000-m rowing ergometer test. A fixed-foot stretcher air-braked Concept2 Model D ergometer equipped with a PM5 monitor (Concept2) was used. Validity and reliability of the Concept2 Model D ergometer have been reported elsewhere.^{3,17} HR was recorded with a Polar M430 HR monitor and an H10 chest sensor (Polar Electro Oy).

Statistical Analyses

Descriptive data are presented as means and SDs. The normal distribution of the data was confirmed by the Shapiro–Wilk test ($P > .05$). A 2-way (muscle group [upper body vs lower body] $\times$ level of effort [high-fatigue vs low-fatigue]) repeated-measures analysis of variance (ANOVA) with Bonferroni post hoc corrections applied to the following dependent variables: T250, T500, and T1000 and the average HR reached during the 1000-m rowing ergometer test. The homogeneity of the variance in the ANOVAs was confirmed by the Levene test ($P > .05$). The same variables were also compared between each RT protocol and the control condition through paired samples t tests and percent differences ($\Delta[\%] = [\text{RT protocol mean value} - \text{Control protocol mean value}] / \text{RT protocol mean value} \times 100$). Statistical analyses were performed using the software package SPSS (version 25.0). Statistical significance was set at $P < .05$.

Results

Descriptive data and the results of the different ANOVAs are presented in Table 1. The ANOVAs applied to the performance variables (T250, T500, and T1000) always revealed a significant main effect of the level of effort due to a lower rowing performance (ie, higher times) observed after the high-fatigue RT protocols ($P \leq .002$), while the main effect of the muscle group never reached

statistical significance ($P \geq .142$). The only significant interaction between the muscle group and level of effort was observed for T1000 ($P = .013$), which was caused by the greater differences between the high- and low-fatigue RT protocols for the lower-body muscles ($\Delta = 5.0$ s) compared with upper-body muscles ($\Delta = 2.1$ s). The average HR obtained during the 1000-m rowing ergometer did not differ between the different RT protocols.

The comparisons of the performance variables (T250, T500, and T1000) and the average HR obtained in the 4 RT protocols with respect to the control condition are depicted in Figure 2. Regardless of the muscle group considered, the 3 performance variables were always significantly deteriorated following the high-fatigue RT protocols compared with the control condition ($P \leq .002$; $\Delta \geq 1.0\%$). However, regardless of the muscle group considered, no significant differences were observed for any performance variable between the low-fatigue RT protocols and the control condition ($P \geq .200$; $\Delta \leq 0.5\%$). The average HR was significantly lower for the control condition compared to the 4 RT protocols ($P \leq .043$; $\Delta \geq 1.0\%$).

Discussion

This study was designed to investigate the potential interference effect on rowing ergometer performance induced by different RT protocols, characterized by variations in the level of effort and muscle groups targeted during the exercises. The main findings revealed that (1) rowing ergometer performance exhibited a decline following completion of high-fatigue RT protocols, while no significant changes were observed after low-fatigue RT protocols, and (2) the impact on rowing ergometer performance was consistent regardless of the muscle groups targeted in the RT protocols. These results emphasize the importance of prioritizing effort level over specific exercises in RT protocols to minimize interference on rowing performance.

The training volume and number of repetitions completed within a set are crucial factors for short-term adaptations in highly trained rowers during CT.^{1,6,18} In this study, rowing ergometer performance (T250, T500, and T1000) was found to be deteriorated only by high-fatigue RT protocols involving both upper- and lower-body exercises. These results support previous findings that moderate training volumes lead to greater short-term improvements in rowing performance compared to high-volume RT protocols.^{6,19–21} The residual fatigue promoted by RT protocols leading to failure could be caused by a reduction in both motor unit recruitment and rapid voluntary neural activations.^{1,7} In fact, it has been suggested that training to failure slows down recovery up

Table 1 Comparison of 1000-m Rowing Ergometer Performance After Completing Different Resistance Training Protocols

Variable	ANOVA			UBHF	UBLF	LBHF	LBLF
	Muscle group	Level of effort	Interaction				
T250, s	$F = 1.2$; $P = .302$	$F = 15.2$; $P = .002$	$F = 0.1$; $P = .816$	53.5 (6.2)	52.5 (6.5) ^{a,b}	53.7 (6.4)	52.7 (6.2)
T500, s	$F = 1.3$; $P = .267$	$F = 21.9$; $P < .001$	$F = 1.5$; $P = .238$	107.8 (12.2)	106.3 (12.8) ^{a,b}	108.5 (12.5)	106.3 (12.2) ^{a,b}
T1000, s	$F = 2.5$; $P = .142$	$F = 75.5$; $P < .001$	$F = 8.4$; $P = .013$	218.3 (25.3)	216.2 (24.4) ^{a,b}	220.6 (24.8)	215.6 (24.5) ^{a,b}
HR, beats min ⁻¹	$F = 1.9$; $P = .190$	$F = 0.1$; $P = .796$	$F = 0.0$; $P = 1.000$	185.9 (5.6)	186.1 (5.2)	186.7 (4.7)	186.9 (4.4)

Abbreviations: ANOVA, analysis of variance; HR, mean heart rate during the 1000-m rowing test; LBHF, lower-body high-fatigue protocol; LBLF, lower-body low-fatigue protocol; T250, time to 250 m; T500, time to 500 m; T1000, time to 1000 m; UBHF, upper-body high-fatigue protocol; UBLF, upper-body low-fatigue protocol. ^aSignificantly lower than UBHF. ^bSignificantly lower than LBHF. Entries in bold type indicate significance of $P < .05$.

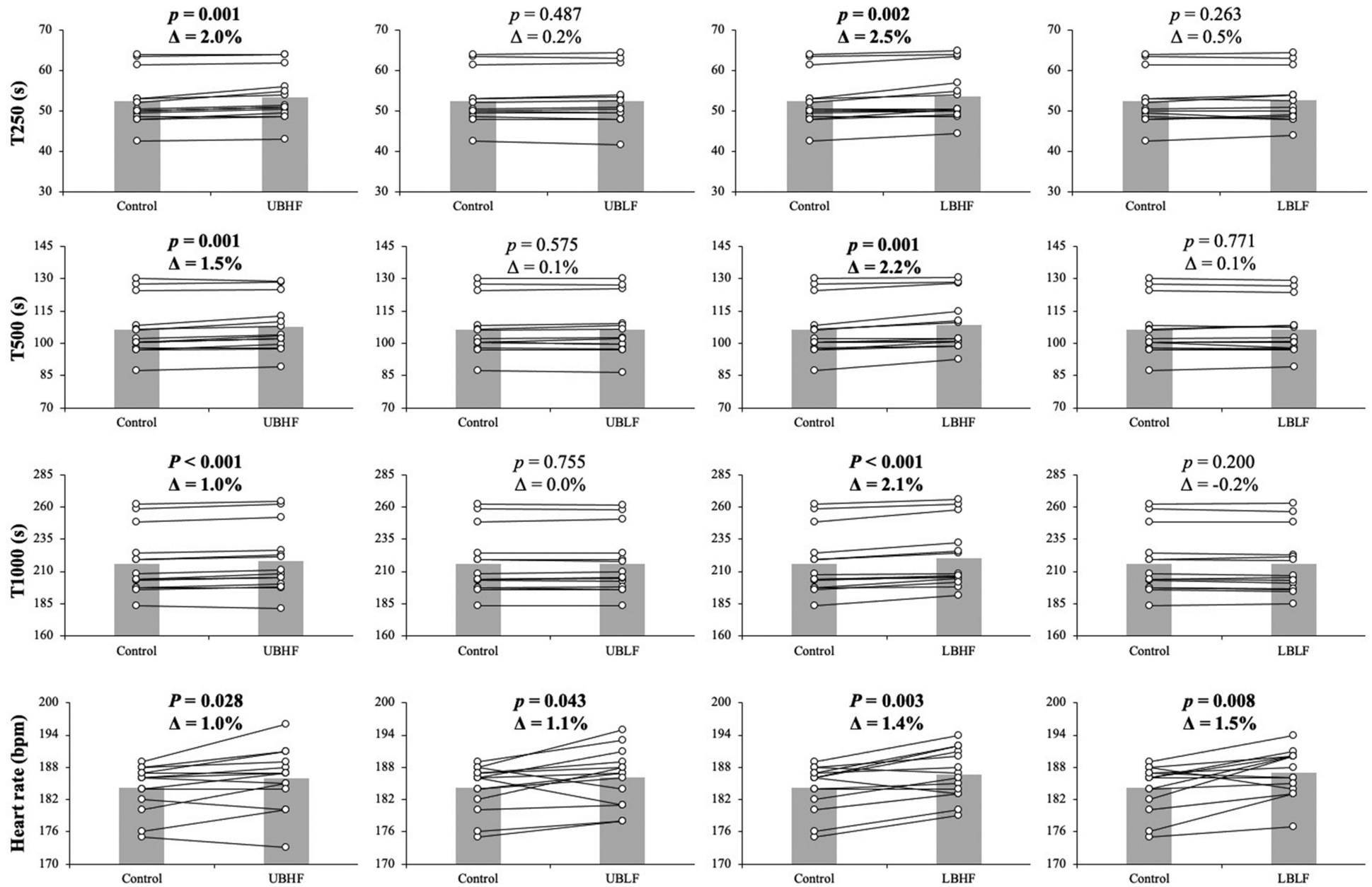


Figure 2 — Comparison of 1000-m rowing ergometer performance between the control condition and the different RT protocols. Heart rate indicates mean heart rate during the 1000-m rowing test; LBHF, lower-body high-fatigue protocol (middle-right panels); LBLF, lower-body low-fatigue protocol (right panels); RT, resistance training; T250, time to 250 m; T500, time to 500 m; T1000, time to 1000 m; UBHF, upper-body high-fatigue protocol (left panels); UBLF, upper-body low-fatigue protocol (middle-left panels). Percentage differences: $\Delta (\%) = (\text{RT protocol mean value} - \text{Control protocol mean value}) / \text{RT protocol mean value} \times 100$. Statistical significance was set at $P < .05$.

to 24 to 48 hours postexercise.¹³ Supporting this notion, Held et al²⁰ demonstrated that halting the sets after surpassing a velocity loss of 10%, in comparison to performing RT sets to failure, accelerated recovery and reduced stress in highly trained male and female rowers. The available scientific evidence strongly supports the notion that, regardless of the relative targeted intensity, rowers should avoid utilizing RT protocols that lead to failure to prevent any negative impact on subsequent rowing performance.

Compelling evidence suggests that strength capacity is positively associated with rowing performance.^{10,14,22} Recent studies by Pérez-Castilla et al²³ and Giroux et al²⁴ have demonstrated that 2000-m rowing ergometer performance is strongly associated with the load–velocity and force–velocity relationship variables obtained during the squat jump and prone bench pull exercises. Similarly, Lawton et al¹⁴ and Huang et al²² revealed that the mechanical performance produced during the leg press and bench pull exercises are powerful predictors of 2000-m rowing performance. However, to our knowledge, this is the first study to directly compare the potential interference effects of upper- and lower-body RT protocols on subsequent rowing performance. Contrary to our hypothesis, which anticipated a greater reduction in rowing performance following a lower-body RT protocol due to the greater involvement of the legs in the specific exercise,¹⁴ our findings revealed comparable changes in rowing ergometer performance regardless of the muscle groups targeted by the RT protocols. These results can be attributed to the fact that, although the legs are more actively engaged than the arms during rowing strokes, it is known that arm muscles tend to be more fatigable than leg muscles due to the larger proportions of fast (type II) fibers, as shown during dynamic contractions at maximal voluntary shortening velocity in the elbow flexor and knee extensor muscles.^{25,26}

The extent to which physiological parameters during rowing can be influenced by prior RT remains poorly understood and insufficiently investigated. Jürimäe et al²⁷ investigated the HR responses of rowers during a rowing ergometer test, bench pull exercise, and leg press exercise, revealing higher mean and maximal HR values in the following order: rowing ergometer exercise, leg press exercise, and bench pull exercise. This is the first study to examine whether performing the leg press and bench pull exercises prior to rowing impacts the subsequent HR response during a 1000-m rowing ergometer test. The average HR during the rowing ergometer test was significantly lower in the control condition compared with the 4 RT protocols; however, no significant differences were found among the RT protocols. It is important to note that HR is a component of late physiological response, and during a rowing ergometer test, the increase in HR may not occur as rapidly as it does following the RT protocols, where HR reaches a working range earlier, potentially explaining the observed differences with the control condition.^{28,29}

While this study offers valuable insights into the interference effect of various RT protocols on rowing ergometer performance, there are certain limitations to consider. First, the study focused on short-term adaptations and did not assess long-term effects, which could provide a more comprehensive understanding of the impact of RT on rowing performance. Second, the sample size consisted solely of male participants with a homogeneous competitive background, which restricts the generalizability of the findings to rowers of different competitive levels or female rowers. Furthermore, this study primarily concentrated on rowing performance and only examined a single physiological parameter (HR), failing to consider other important factors such as technique, coordination, or biomechanical aspects that could also contribute to rowing

performance. A more comprehensive assessment of these factors would lead to a more holistic understanding of the interference effect of RT protocols performed prior to rowing. Addressing these limitations in future research will provide a more robust foundation for understanding the relationship between RT and rowing performance.

Practical Applications

Rowers can confidently integrate the leg press and bench pull exercises into their training routine before rowing practice. However, to avoid compromising rowing performance, sets should be performed with moderate levels of effort. By incorporating these exercises with moderate effort, rowers can effectively enhance overall strength and conditioning without negatively impacting their rowing performance. This approach optimizes their training regimen while maintaining peak performance during rowing practice.

Conclusions

High-fatigue resistance-training (RT) protocols that led to failure had a detrimental effect on subsequent rowing ergometer performance, while low-fatigue protocols showed no significant impact. The specific muscle groups targeted in the RT protocols did not influence rowing performance. Additionally, all RT protocols resulted in similar increases in heart rate during rowing compared with not performing any RT protocol prior to the test. These findings highlight the importance of prioritizing the level of effort over specific muscle groups in RT protocols to minimize interference on rowing performance. Coaches and rowers can utilize this information to design training programs to optimize performance by avoiding high-fatigue RT protocols that lead to failure before rowing practice.

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